Minor Project Guidelines – EDA

Note:

Students are required to complete **at least one project** from the three assigned minor projects. These projects are designed to integrate concepts from **Basic Python**, **Pandas**, **Exploratory Data Analysis (EDA)**, and **Data Visualization** using **Matplotlib** and **Seaborn**.

Submission Deadline

The final submission deadline is **Monday, End of Day (EOD)**.

Development Environment

Students may use any of the following tools to complete the project:

1. Google Colab
2. Jupyter Notebook
3. Visual Studio Code (VS Code)

Project Requirements

Each submission must include:

1. A clear and well-defined problem statement
2. Well-structured, readable, and logically organized code
3. Proper use of **Pandas** for data handling and manipulation
4. **Exploratory Data Analysis (EDA)** with meaningful insights
5. Appropriate visualizations using **Matplotlib** and **Seaborn**
6. Explanatory comments and markdown cells wherever applicable

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Minor Project 1: Sales Performance Analysis System

Dataset

`product_sales_dataset_final.csv`

Problem Statement

Organizations collect daily sales data but often lack a clear understanding of overall performance, trends, and category-wise contributions.
This project focuses on analyzing sales data to extract meaningful insights.

Objectives

- Read and process a structured sales dataset
- Perform data cleaning and validation
- Conduct exploratory data analysis to understand sales behavior
- Visualize trends and comparisons using charts

Key Concepts and Functions Used

Core Python

- Variables and data types
- Conditional statements
- Loops
- Functions

Pandas

- `pd.read_csv()`

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- `df.head()`, `df.info()`, `df.describe()`
- `df.isnull()`, `df.fillna()`
- `df.groupby()`
- `df.sort_values()`

EDA

- Summary statistics
- Category-wise aggregation
- Trend identification

Visualization (Matplotlib & Seaborn)

- Line plots for sales trends
- Bar charts for category-wise sales
- Count plots for transaction distribution

Logical Flow of Implementation

1. Load the sales dataset using Pandas
2. Inspect structure, data types, and missing values
3. Clean the dataset (handle null values and correct data types)
4. Perform EDA to analyze total sales, average sales, and category performance
5. Visualize insights using appropriate plots
6. Summarize key findings

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Minor Project 2: Student Performance and Attendance Analyzer

Dataset

`Student Academic Performance.xlsx`

Problem Statement

Educational institutions need to analyze student performance and attendance to identify patterns, weak areas, and overall outcomes.
This project analyzes academic records to support data-driven decisions.

Objectives

- Analyze student marks and attendance data
- Identify performance trends and distributions
- Compare subject-wise and attendance-based performance
- Visualize academic insights

Key Concepts and Functions Used

Core Python

- Lists and dictionaries
- Conditional logic
- User-defined functions

Pandas

- `pd.read_csv()`
- `df.shape`, `df.columns`, `df.dtypes`

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- `df.isnull()`, `df.fillna()`
- `df.groupby()`
- `df.value_counts()`

EDA

- Distribution of marks
- Attendance vs performance analysis
- Subject-wise comparisons

Visualization (Matplotlib & Seaborn)

- Histograms for marks distribution
- Box plots for subject-wise performance
- Bar plots for attendance categories

Logical Flow of Implementation

1. Load the student dataset
2. Inspect and clean the data
3. Analyze overall performance statistics
4. Perform group-based analysis (by subject or attendance)
5. Visualize distributions and comparisons
6. Interpret results and academic insights

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Minor Project 3: Monthly Expense and Spending Pattern Analysis

Dataset

`SuperMarket Analysis.csv`

Problem Statement

Individuals often track expenses but struggle to understand spending patterns and control unnecessary costs.
This project focuses on analyzing personal expense data to identify trends and spending behavior.

Objectives

- Analyze monthly expense data
- Identify high-spending categories
- Understand daily and category-wise expense patterns
- Visualize spending behavior

Key Concepts and Functions Used

Core Python

- File handling basics (optional context)
- Conditional statements
- Functions

Pandas

- `pd.read_csv()`
- `df.head()`, `df.info()`, `df.describe()`

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- `df.isnull()`, `df.fillna()`
- `df.groupby()`
- Basic date-related operations

EDA

- Category-wise expense analysis
- Trend analysis over time
- Outlier identification

Visualization (Matplotlib & Seaborn)

- Line plots for expense trends
- Pie charts or bar charts for category-wise spending
- Box plots for expense variability

Logical Flow of Implementation

1. Load the expense dataset
2. Inspect data structure and missing values
3. Clean and preprocess the dataset
4. Analyze total and average expenses
5. Visualize spending patterns and trends
6. Draw conclusions and recommendations

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Expected Learning Outcomes

By completing these projects, students will be able to:

- Apply Python fundamentals in real-world data scenarios
- Use Pandas for data cleaning, manipulation, and aggregation
- Perform Exploratory Data Analysis (EDA) systematically
- Create meaningful visualizations using Matplotlib and Seaborn
- Develop structured, end-to-end analytical workflows

Submission Instructions

1. Submit the project as a `.ipynb` **notebook file**
2. Ensure **all cells are executed** before submission
3. Include all **visual outputs** in the notebook
4. Use **meaningful variable names** and a clear structure

Late submissions may not be considered unless prior approval is obtained.
Evaluation will be based on **correctness, clarity, analytical thinking, and presentation quality**.

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